MIND THE GAP

Deciphering the Trade-off Between Transit Accessibility and Real-Estate Value

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DOMAIN

This project explores how public transportation affects real-estate prices in Jerusalem. Usually, the regular intuition of people is that being close to a station is a good thing. But we want to find the exact "Sweet Spot"- the perfect distance where an apartment has easy access to transit, without suffering from the noise and crowds of a busy station. In addition, we look at urban planning. Using our data, we suggest the best locations to build new stations for neighborhoods that currently have poor public transit access.

DATA

1st Data Source: Public Transportation Stations Dataset, it contains 2404 records and comes in a csv file form. We acquired it from the Data.gov.il site.

2nd Data Source: Jerusalem Real Estate Transactions, the size of the dataset is 2.33 MB and it contains residential real estate listings from Yad2 (of early 2026). We acquired it from Kaggle

3rd Data Source: Transit Trip Frequency Data. The size of the dataset is 932.4 MB. It comes in a form of a TXT file and contains the full schedule of buses per stations. We acquired it from the Ministry of Transport, it's a GTFS file.

We did encounter some biases while handling the datasets.

- "Fake Stations"- The raw stations dataset contains every single bus stop in the city, including abandoned stops or stops that only see one bus a day. Treating these as equal to a major central station would severely bias our distance calculation. To solve this, we merged the stations with the GTFS file and filled missing frequencies with zero. Then, we used the DBSCAN algorithm to cluster only the highly active stations.
- Temporal Sampling Bias- The Yad2 dataset is a snapshot of active listings. It is not a complete dataset representing all the real-estate prices in Jerusalem. This means our dataset is a sample of the market and not the entire population. We acknowledged this bias. While it means the absolute number of apartments we count in our Urban Optimization is smaller than reality, it serves as a highly accurate representative sample.

PROBLEM 1- FINDING THE TRANSIT "SWEET SPOT"

- Description- How does the physical distance to a busy public transportation hub impact the price per square meter of residential properties in Jerusalem? Is there an optimal distance- a "Sweet Spot"- where the property value peaks? **Our hypothesis** was that the relationship between property price and transit is not a simple line. We expected prices to be relatively low very close to a hub (due to noise, pollution, crowds), rise to peak at a comfortable walking distance and then gradually decrease as the distance grows and access becomes inconvenient.

- Solution- We combined real estate data with public transportation schedules.
  - Step 1- Finding Transit Hubs using **DBSCAN** to identify major transit hubs from thousands of small bus stops. We preferred using DBSCAN over K-Means because we didn't know how many hubs, we will end up with in the end.
    - Parameters- We used Haversine distance metric which calculates the shortest great-circle distance between two points on a sphere (like earth, in our case). We set **eps** to be 300 meters as the standard walking radius around a central station. We set **min_samples** = 5, to ensure we only capture major, busy hubs with multiple lines, filtering out small, single line stops. We also set **freq_threshold**=400 to filter out stations with less than 400 daily bus visits to focus only on active areas.

    The visualization demonstrates the effectiveness of DBSCAN. The algorithm successfully identified distinct, dense transit hubs while correctly classifying highly active but geographically isolated bus stops as 'noise'.

  - Step 2- Calculating distances. Once we had the exact coordinates of the active transit hubs, we used the Haversine Formula to calculate the exact physical distance between each apartment and its nearest hub.
  - Step 3- Finding the Sweet Spot using a decision tree regressor. We trained a supervised machine learning model to predict the price per square meter based on the distance to the hub. We used this method since this model can capture non-linear relationships and thresholds.
    - Parameters- we set **max_depth** = 4 and **min_sample_leaf** = 40. We limited the depth to prevent overfitting to the noisy real estate data. Setting min_sample_leaf = 40 ensures that every prediction is backed by a solid group of at least 40 apartments, representing a true market trend.
- Evaluation
  - Evaluation Criteria- Success in this problem is defined by our model's ability to clearly identify and visualize a non-linear relationship between distance to transit and property prices. The project will be considered "Successful" if we can prove the existence of "Sweet Spot"- peak in price at a medium distance, surrounded by lower prices at extreme close or far distances. Our ground truth here was the actual real-estate prices of apartments in Jerusalem. We evaluate our decision tree regressor against a baseline linear regression model. Since linear regression can only draw a straight line, it assumes that distance has a constant positive or

negative effect. If our decision tree significantly out preforms the baseline, we consider our model successful. To ensure out finding are not random noise, we performed a Train/Test split and evaluated the model's accuracy on unseen data. Also, we ensured that every data we acquired will be backed with at least 40 other real apartments (min_sample_leaf).

- Setup- we first merged out cleaned real estate data with our DBSCAN defined hubs and calculated the Haversine distance for each apartment to its nearest hub. To avoid biases, we removed extreme price outliers (such as luxury mansions and villas skewing the model's understanding of the market) and applied a geographic filter to include only properties within 3000m of a hub.
- Results- We did in fact proved the existence of the "Sweet Spot". The baseline linear regression model failed to capture the complexity of the market, showing only a weak, flat trend. However, our Decision Tree successfully identified distinct distance thresholds.
- Visualization- This plot highlights the non-linear relationship between property values and distance to transit hubs. The baseline Linear Regression completely misses the market. In contrast, out decision tree successfully captures the market's true complexity. In the proximity zone (~500m) the prices are relatively high (indicating the importance of close transit. Afterwards, we see a decrease that continues until about ~1500m- the luxury peak indicating the prices increase when distancing enough from crowds and transit, but not too much. And at about ~2000m we see a flat line indicating the distance is indeed affecting the real estate prices when you're too far from transit.
- Visualization 2- This boxplot visualizes the median property prices across out newly defined five distance zones. It confirms the broad macro-economic trend that greater distance from transit generally lowers property valuations. However, we can still see the "Luxury Peak", which exhibits a notably wide price variance and a higher median price compared to the adjacent Transit Periphery.

PROBLEM 2- OPTIMIZING TRANSIT COVERAGE (SOLVING "TRANSIT DESERTS")

- Problem description- Jerusalem has many residentials areas where public transportation is non-existent or too far. This creates "Transit Deserts". If the municipality of Jerusalem wanted to open a fixed number of transit stations for the public, where will it be most convenient and how do we optimize the location so it will cover as much "deserts"?

- Solution- to solve this, we implemented a greedy algorithm designed for maximum coverage. We treated every unserved property (distant for more than 800 m) as a potential candidate for a new station. The algorithm works iteratively: in each step, it places a new hub at a specific location that covers the highest number of previously unserved properties. A greedy approach is ideal here since we want to cover as much properties in Jerusalem that suffer the transit problem. We set a radius of 800 meters, which represent the standard "walkable" distance for public transit commuters.
- Evaluation
    - Evaluation Criteria- Our success is defined by "Marginal Coverage Gain". We measure how many additional properties are brought withing the 800-meter radius by each new station.
    - Setup- We ran the algorithm for 3 iterations (Simulating 3 new transit hubs). We avoided biases by calculating distances using the Haversine formula, which accurately accounts for the Earth's curve, ensuring our coverage radius is geographically correct.
    - Results- we generated a map displaying- existing served properties (in gray), transit deserts (in coral) and the optimal places for 3 new hubs (as stars).
    - Visualization- This map illustrates our data-driven approach to solving Jerusalem's public transportation gaps. It displays properties already served by existing transit in gray, while highlighting unserved "transit deserts" in coral. The stars represent the three new transit hubs strategically placed by our greedy algorithm. By targeting the densest clusters of unserved properties, the algorithm ensures maximum coverage and efficiency for the city's new infrastructure budget.
    - Visualization 2- this bar chart quantifies the success of our greedy algorithm in addressing Jerusalem's "Transit Deserts". It displays the exact number of new, previously unserved properties that gain transit access with each strategically placed hub.

FUTURE WORK

Although we modeled the "Transit Sweet Spot" and identified optimal locations for new hubs, there are several ways to evolve this project-

- Adding Socio-economic factors- we could try and improve the results of this project by considering the socio-economic factor, such as average income per neighborhood or household car ownership rates. This would help us prioritize new transit hubs in neighborhoods where residents depend on public transportation the most.
- Future plannings- Our current model focuses on bus transit hubs. Future iteration could incorporate the Jerusalem light rail system and future developments of its infrastructure which is well planned for upcoming years.

CONCLUSION

This project demonstrated that the relationship between property value and transit accessibility is not linear, but rather a “Sweet Spot” curve. Through spatial data analysis and machine learning, we proved that there is an optimal distance from transit hubs that maximize apartment value while minimizing the negative impacts of noise and congestion. Our prescriptive urban optimization model showed that a simple greedy algorithm can identify “Transit Deserts” and suggest highly effective locations for new transit hubs. We hope this model can serve as an example for future planning in our loved city Jerusalem.